



The Exeter Ship Canal reconnected the English inland port of Exeter to the sea, bypassing the section of the Exe River that was no longer navigable.

BEGUN IN 1564, the Exeter Ship Canal was completed in 1566 or 1567. A man-made **navigable channel**, it reestablished the town of Exeter in England as a port after centuries of blockages on the Exe River by weirs constructed to power watermills. Probably the UK's first artificial waterway, the Exeter Canal was a forerunner of the prolific canal-building that came with the Industrial Revolution in the 18th century.



“ HE DOES SMILE HIS FACE INTO MORE LINES THAN IS **IN THE NEW MAP** WITH THE AUGMENTATION OF THE INDIES. ”

William Shakespeare, English playwright, *Twelfth Night*, c.1602

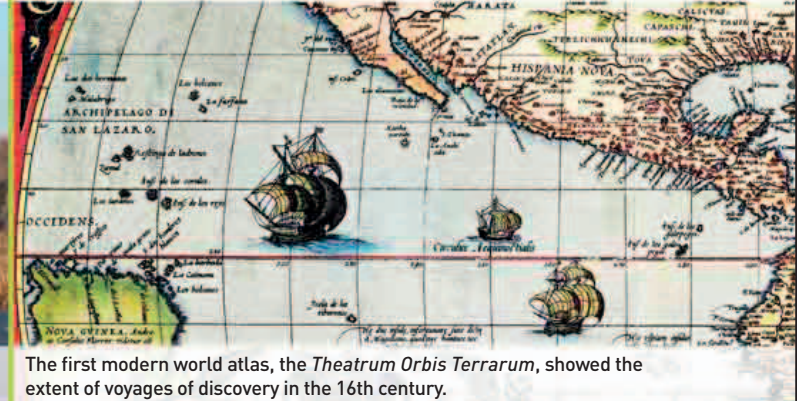
Maps became increasingly important during the 16th century, as traders and explorers made voyages around the world. While cartographers could make accurate globes showing oceans and continents,

these were not convenient for navigation. The problem was how to represent the three-dimensional Earth in two dimensions. In 1569, Flemish cartographer **Gerardus Mercator** (1512–94), already famous for his globes and maps of Europe, devised a new way to represent the

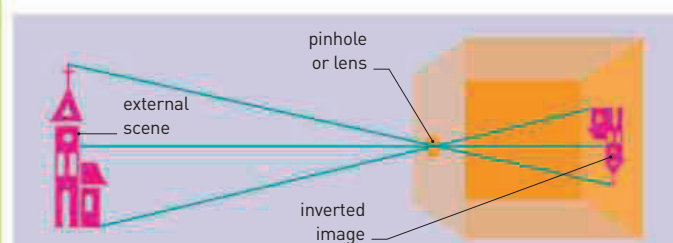
Gerardus Mercator Mercator's projection, a method of showing the spherical world as a two-dimensional map, enabled accurate nautical navigation and still remains in use.

curved surface of Earth when designing his paper **map of the world**. Now known as **Mercator's projection**, this method shows the lines of longitude as equally spaced, parallel, vertical straight lines, and the lines of latitude as perpendicular to them—as if they were projected onto a cylinder enclosing a globe. Although this projection distorted the shapes and sizes of large land masses and oceans, it was particularly useful for navigation, since compass courses could be shown as straight lines.

Although the German–Swiss alchemist and physician **Paracelsus** (1493–1541) was a prolific writer and notorious self-publicist, few of his works were published during his lifetime. One of the most influential, *Archidoxa*, was published posthumously in Kraków in 1569. This **rejected the magical elements of alchemy** and was important in the development of modern chemistry and medicine.



The first modern world atlas, the *Theatrum Orbis Terrarum*, showed the extent of voyages of discovery in the 16th century.



CAMERA OBSCURA

A camera obscura (Latin for “dark chamber”) is a simple room or box with a small hole in one of its walls. Light from outside passes through the hole and falls onto the inside of the opposite wall, projecting an image of the scene outside. The image, which appears upside down, is sharper with a smaller pinhole, or brighter with a larger hole, and can be focused by adding a lens.

ENCOURAGED BY HIS FRIEND AND COLLEAGUE MERCATOR, the Flemish cartographer **Abraham Ortelius** (1527–98) published his *Theatrum Orbis Terrarum* (*Theater of the World*) in 1570. He had previously made a large map of the world on eight sheets, and several separate maps of various parts of the world, but this collection of 53 maps and accompanying text in book form was the **first modern world atlas**—a term suggested by Mercator. After its original publication in Latin, it was translated into several other languages, and in these later editions Ortelius added several more maps and corrected inaccuracies.

Italian polymath **Giambattista della Porta** (c.1535–1615) first published his book *Magiae Naturalis* (*Natural Magic*) with descriptions and observations of all the sciences in 1558, but it proved so popular that it was revised and expanded in a

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THE NUMBER OF MAPS IN THE FIRST WORLD ATLAS

1566/67 The Exeter Ship Canal is completed

1569 Paracelsus's treatise on alchemy and medicine, *Archidoxa*, is published posthumously

1570 Abraham Ortelius publishes the first modern atlas, the *Theatrum Orbis Terrarum*

1569 Gerardus Mercator publishes a complete map of the world

c.1570 Giambattista della Porta describes an improved camera obscura